

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Semester : VIII

Branch : Biomedical Engineering / Electronics Engineering / Electronics and Telecommunication Engineering / Electronics and Telecommunication Engineering (Sandwich)/ Electronics and Communication Engineering

Subject Code & Name: BTETPE802D Industrial Automation and Control

Max Marks: 60

Date:17-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Give a guideline for selection of controller mode in PID controller?	(L1)	6
B) Explain Data Acquisition System with the help of its block diagram?	(L2)	6
C) Give the Advantages of Hydraulic Actuation Systems.	(L2)	6
Q.2 Solve Any Two of the following.		12
A) What are features of the standard that promotes its versatility and efficiency in programming control tasks?	(L3)	6
B) Develop the transfer function of the feedforward-feedback controller?	(L1)	6
C) Give the advantages and disadvantages of cascade control?	(L3)	6
Q. 3 Solve Any Two of the following.		12
A) What are the different types of Automation System?	(L1)	6
B) Discuss with block diagram the Elements of a measuring system.	(L2)	6
C) List and explain Process control input and outputs in stamping process?	(L3)	6
Q.4 Solve Any Two of the following.		12
A) How CNC machine tool systems can be classified and give the advantages of CNC machines?	(L2)	6
B) Explain major aspects of production planning?	(L1)	6
C) Explain the Architecture of PLC with the help of block diagram.	(L2)	6

Q. 5	Solve Any Two of the following.		12
A)	Explain the different layers of Automation Pyramid with the help of its Block diagram?	(L1)	6
B)	Which activities are include in the scope of production planning?	(L3)	6
C)	Explain the static characteristics of Measurement Systems?	(L2)	6

***** End *****